

When should VEGO-AI involve a human?

A visual baseline from existing AirTravel evidence

WHAT WAS STUDIED

We can now see where an alert can be raised. We still cannot say whether the alert is correct.

Executive Summary

- VEGO-AI ON adds agent coordination, an inspectable Q&A log, and Detector-v1 labels.
- OFF is a direct workflow. It finished all cases with less time and cost in this run.
- Human labels are still missing. Quality, accuracy, benefit, and workload reduction are not measured.

Public external LLM-generated AirTravel corpus 12 seeded candidate-model artifacts from 21 eligible gpt-5.6-luna · PROVIDER_DEFAULT_NOT_SENT · one day Not student data · not representative

3 inputs overlap Study 1 · no outputs reused or pooled

Provider-default stochastic run; a second run may differ.

Evidence ladder

ARCHIVAL-RETROSPECTIVE DESCRIPTIVE EVIDENCE

1 · Archival Study 1

4 purposive cases · 3 complete episodes · 3/3 candidate labels

ENGINEERING-ONLY FIXTURE

ENGINEERING_FIXTURE_NOT_SCIENTIFIC

2 · Engineering checks

9/9 rule modes · 724 fixture calls · 0 provider calls

DESCRIPTIVE_INSTRUMENT_ROBUSTNESS_ON_ONE_ACCEPTED_RUN
500/500 seeded permutations · 3/3 named reorderings

PROSPECTIVE EMPIRICAL EVIDENCE

3 · Prospective comparison

12 matched AirTravel cases · ON versus OFF · ON STOPPED_AT_CAP

NOT_MEASURED

4 · Human validation

0 raters scored · alert correctness remains open

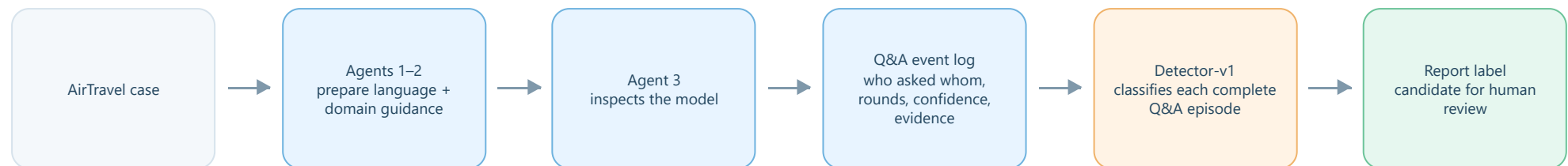
Run S2P-LIVE-20260908T192135Z ON = STOPPED_AT_CAP N = 12 planned cases

What changes when VEGO-AI is ON?

Same input, two different workflow packages

VEGO-AI ON

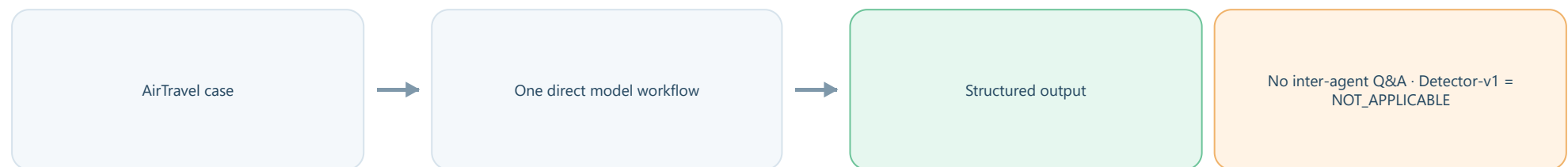
PROSPECTIVE EMPIRICAL EVIDENCE · N=12



Agent 4 — Prospective: 0 Agent-4-labelled requests before cap · separate archival path: EXECUTED_THEN_BLOCKED · queue NOT_AVAILABLE

VEGO-AI OFF

BASELINE · N=12



VEGO-AI added

- ✓ observable coordination
- ✓ traceable Q&A episodes
- ✓ automatic reporting labels

It did not add

- ✗ automatic correction
- ✗ an Agent-4 queue in this capped run
- ✗ proof of better quality

Which logs mean what

qa_events.jsonl · Detector source when validated

user_actions.log · GUI audit; not Q&A evidence

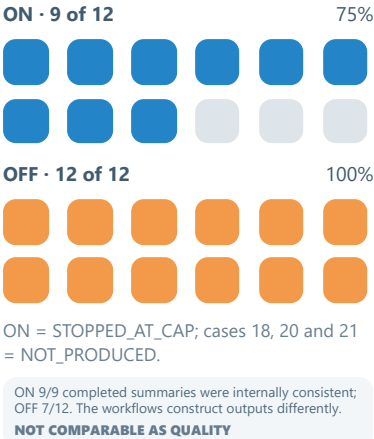
interaction_log.json · possible model trace; not Detector evidence

Agent-4 output · separate variability mechanism

Operational baseline: ON versus OFF

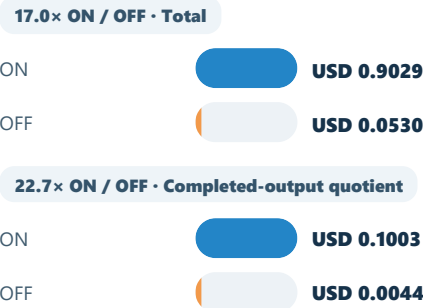
Descriptive measurements on 12 planned cases

Completed outputs



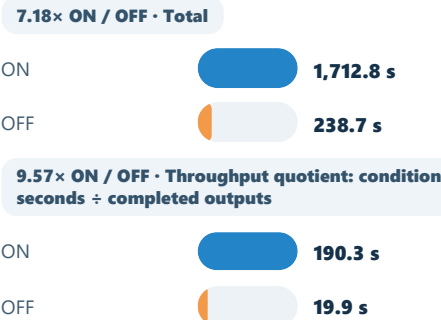
Source: analysis.json · **Condition:** ON + OFF
Denominator: 12 planned cases per condition
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: completed output artifacts ÷ planned cases
Limitation: ON stopped at cap; completion is not quality.

Recorded cost



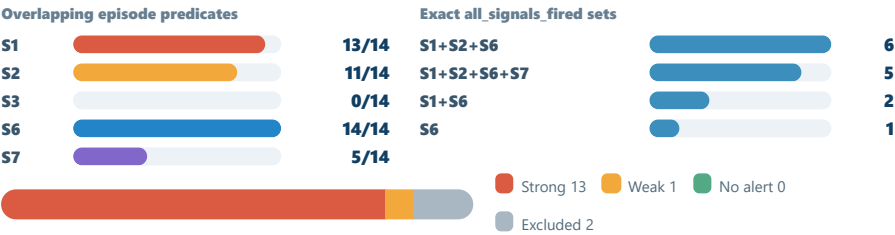
Source: analysis.json · **Condition:** ON + OFF
Denominator: all recorded requests; quotient uses ON 9 / OFF 12 completed outputs
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: recorded USD; quotient = condition cost ÷ completed outputs
Limitation: ON includes 2 shared setting-level requests and concurrency attributions; not isolated per-case cost or quality.

Elapsed time



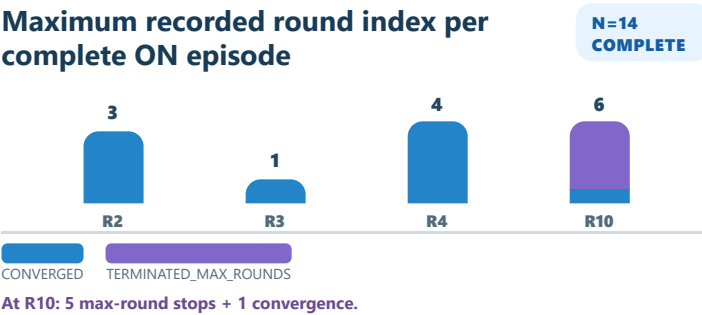
Source: analysis.json · **Condition:** ON + OFF
Denominator: whole conditions; quotient uses ON 9 / OFF 12 completed outputs
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: condition wall-clock; throughput quotient = seconds ÷ completed outputs
Limitation: Not mean per-case latency; ON ran concurrently, with one model on one day.

Detector-v1 predicate occurrence and combinations



Source: public-aggregate.json · **Condition:** VEGO-AI ON
Denominator: 16 recorded episodes for classes; 14 complete episodes for predicates
Metric: overlapping predicate presence and exact signal set per episode
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Limitation: Model self-report plus structural fields; signals may co-occur and are not calibrated uncertainty or human judgment.

Maximum recorded round index per complete ON episode



Source: public-aggregate.json · **Condition:** VEGO-AI ON
Denominator: 14 complete Q&A episodes; 2 incomplete technical episodes excluded
Metric: maximum recorded round_index per complete episode
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Limitation: Process state—not difficulty, failure, correctness, quality, or human workload.

Where do Detector signals appear—and when can a human enter?

Observed process evidence, reporting point, and the missing human judgment

What is observed—and what remains unresolved

No objective known/unknown field is recorded **NOT_MEASURED**

Detector-v1 uses answer-confidence fields reported by the model plus structural and control fields.

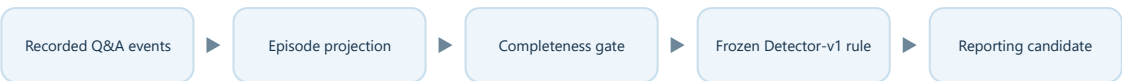
S1 Low · S2 Medium (model self-report) · S3 no evidence reference · S6 > 1 round · S7 max-round termination

model self-report not correctness NOT_AVAILABLE_IN_TRACKEDAggregate

NOT_MEASURED / NOT_AVAILABLE is an evidence status—not objective absence or calibrated uncertainty.

Where the reporting label is produced

POST_EPISODE_TERMINATION_REPORTING



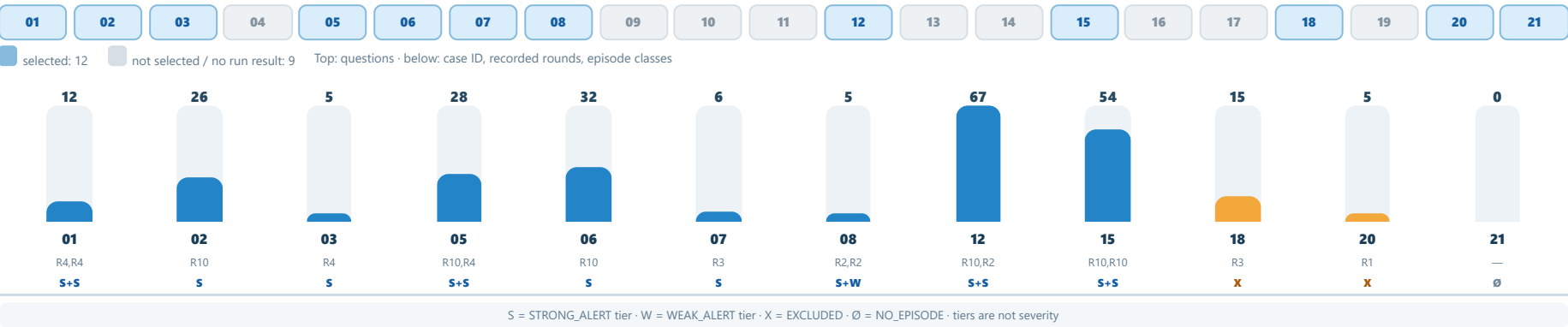
STRONG = S1 OR S3 OR S7 · WEAK = no strong signal AND (S2 OR S6)

Alert = a candidate for human review in the report

POST_EPISODE_TERMINATION_REPORTING · no interruption, correction, or queue write

Q&A volume by selected AirTravel case

One AirTravel task · 21 eligible artifacts · 12 seeded · 9 not selected



Source: public-aggregate.json + case-selection-manifest.json · Condition: VEGO-AI ON
Denominator: 21 eligible artifacts; bars show the 12 seeded cases
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: recorded questions per selected case; badges preserve every episode class
Limitation: Twelve artifacts from one task, not 12 use cases; frequency is not risk, error, importance, or intervention need.

Requested answer-level cuts not exposed by the tracked aggregate

NOT_AVAILABLE_IN_TRACKEDAggregate

Confidence histogram · evidence-reference lengths · first Low-confidence round

When should a human enter?

After the reporting candidate, raters must decide REVIEW_WORTHY / NOT_REVIEW_WORTHY / INSUFFICIENT_INFORMATION. Human outcome = NOT_MEASURED (0 raters).

Before scoring: current pack = 2 raters; supervisor freeze = 3. Reconcile this and adjudicate disagreements.

HUMAN OUTCOME
NOT_MEASURED · 0

Observed Q&A routes: asking Agent 3 · answering Agent 1 = 64 Q · answering Agent 2 = 191 Q · n=255 questions

Separate Agent-4 mechanism: EXECUTED_THEN_BLOCKED · queue NOT_AVAILABLE

Current result: an inspectable escalation baseline—not validated human-intervention detection.